Network flows solutions

A09-F01

The forward uv channel has 2 and the reverse-of-vu channel has 1. Aggregate capacity is 3. One valid allocation cancels 1 on vu and adds 2 on uv. Adding all 3 to uv would violate $5+3 \leq 7$.

A09-F02

P_3=s,y,x,t has residuals 6,3,4 and $\delta=3$. The flow vector is(3,7,3,0,7,3,7). At x,3=0+3; at y,7+0=7. Value rises from 7 to 10.

A09-F03

Sum net outflow at vertices in S. Internal edges cancel, and all vertices except s conserve flow, leaving |f|. Crossing flow equals forward flow minus return flow, at most forward flow, at most forward capacity. Return capacity permits flow but does not force return flow, so it cannot be subtracted.

A09-F04

The reachable set is{s,w,x,y}; outgoing cut edges yz and xt have capacity 7+4=11. The feasible value 11 meets this upper bound. For S={s,w}, outgoing edges sy and wx have capacity 10+9=19, so this cut is not minimum.

A09-F05

Without a path, use residual reachable S. Every S→T edge is saturated; every T→S edge has zero flow, or its reverse residual would cross the cut. Thus equality holds. Conversely equality attains a global upper bound; an augmenting path would exceed it, a contradiction.

A09-F06

There are 16 rounds and the optimum is 16. U needsΘ(log U) bits, so 2 U rounds can be exponential in bit length. Scaling rational capacities to integers proves termination; arbitrary real capacities with arbitrary choices may not terminate.

A09-F07

The new arc is generated by augmentation along $v \to u$. Every prefix of the old BFS augmenting path is shortest, giving old $d(u)=d(v)+1$. Choose the first decreased vertex v on a new shortest path; its predecessor u has not decreased. An old arc contradicts the old triangle inequality; a new arc gives $d(v) \leq d'(v)-2$, also a contradiction.

A09-F08

Each round saturates a critical(u,v), which disappears. Reappearance requires use of(v,u), giving new $d(u)=$new $d(v)+1 \geq$old $d(v)+1=$old $d(u)+2$. Finite distances are at most $V-1$. Thus each direction has $O(V)$ events, $O(E)$ directions give $O(VE)$ rounds, and BFS plus update costs $O(E)$ on the relevant subgraph. Total $O(VE^2)$, plus initialization. Finite nonnegative real capacities are allowed under exact arithmetic; bit costs are separate.

A09-P01

No. a and b have excess 6 and 4, so conservation fails. The active set is $\{a,b\}$. The difference between source outflow 10 and sink inflow 0 is the 10 units stored internally.

A09-P02

$h(a)=h(c)=0$ violates $h(a)=h(c)+1$. Residual neighbors s,c,t have heights 5,0,0, so a relabels to 1.

A09-P03

$\delta=\min(2,6)=2$. Cancel two units on original edge sa, reducing $f(sa)$ from 6 to 4. $e(s)$ increases from -10 to -8 and $e(a)$ decreases from 2 to 0. The source is never active and never relabeled, so $h(s)=5$.

A09-P04

Skip s,c,t to NIL; relabel $0 \to 1$ and reset to s. Skip s and push 3 on ac, keeping current=c. Only on the next iteration does saturation cause an advance to t. Push 1 on at, keep current=t, then advance to NIL on the next iteration. Relabel $1 \to 6$ and reset to s. Return 2 to s and finish. The full operation table follows.

A09-P05

After step 25 the discharge order is a,c,b,a. The first a has zero excess. c relabels and moves from [b,a,c] to [c,b,a]. b has zero excess. The final a returns one unit without relabeling, so the list stays [c,b,a]. An active-only container is no longer the CLRS list of all intermediate vertices. Its list invariant and end-of-list termination proof must be replaced.

A09-P06

A simple residual path has length $k \leq n-1$. Summing height constraints gives $h(s) \leq h(t)+k$, hence $n \leq k$, a contradiction. A preflow can store excess internally. Only after all intermediate excesses vanish is it a feasible flow, to which the no-augmenting-path maximum-flow theorem applies.

A09-P07

Every active vertex has a simple residual path to s, bounding height by $2n-1$. Strict increases give $O(n)$ relabels per vertex and $O(n^2)$ total. Repeated saturation in one direction requires a reverse push and then a height increase of at least two, giving $O(nm)$ saturating pushes. With $\Phi=\Sigma$active h, nonsaturating pushes reduce potential by at least one. Relabels and saturating pushes add $O(n^3+n^2m)$ total. Remove isolated components excluding s/t to obtain $n=O(m)$, giving $O(n^2m)$. Each relabel scans neighbors to find the minimum, costing $O(\deg(u))$, and $O(nm)$ overall.

A09-P08

Final flows are sa=3, sb=2, ac=2, at=1, bc=2, ct=4. The cut S={s,a,b,c} has outgoing edges at and ct of total capacity 5, equal to the flow. $O(n^2)$ relabels define $O(n^2)$ phases, each with $O(n)$ discharge calls, hence $O(n^3)$ calls. A nonsaturating push empties u and immediately ends its discharge, so there are at most $O(n^3)$ such pushes. All other work totals $O(nm)$, giving $O(n^3)$.

Complete operation trace

Each row is the state after the operation. Current entries are in a,b,c order; heights and excesses are in s,a,b,c,t order. NIL means the neighbor scan reached the end.

step	operation	h(s,a,b,c,t)	e(s,a,b,c,t)	current(a,b,c)	L
0	initialize	5,0,0,0,0	-10,6,4,0,0	s,s,a	a,b,c
1	discharge-start a	5,0,0,0,0	-10,6,4,0,0	s,s,a	a,b,c
2	advance a a→s	5,0,0,0,0	-10,6,4,0,0	c,s,a	a,b,c
3	advance a a→c	5,0,0,0,0	-10,6,4,0,0	t,s,a	a,b,c
4	advance a a→t	5,0,0,0,0	-10,6,4,0,0	NIL,s,a	a,b,c
5	relabel-reset a	5,1,0,0,0	-10,6,4,0,0	s,s,a	a,b,c
6	advance a a→s	5,1,0,0,0	-10,6,4,0,0	c,s,a	a,b,c
7	push a a→c δ=3	5,1,0,0,0	-10,3,4,3,0	c,s,a	a,b,c
8	advance a a→c	5,1,0,0,0	-10,3,4,3,0	t,s,a	a,b,c
9	push a a→t δ=1	5,1,0,0,0	-10,2,4,3,1	t,s,a	a,b,c
10	advance a a→t	5,1,0,0,0	-10,2,4,3,1	NIL,s,a	a,b,c
11	relabel-reset a	5,6,0,0,0	-10,2,4,3,1	s,s,a	a,b,c
12	push a a→s δ=2	5,6,0,0,0	-8,0,4,3,1	s,s,a	a,b,c
13	discharge-end a	5,6,0,0,0	-8,0,4,3,1	s,s,a	a,b,c
14	move-front a	5,6,0,0,0	-8,0,4,3,1	s,s,a	a,b,c
15	discharge-start b	5,6,0,0,0	-8,0,4,3,1	s,s,a	a,b,c
16	advance b b→s	5,6,0,0,0	-8,0,4,3,1	s,c,a	a,b,c
17	advance b b→c	5,6,0,0,0	-8,0,4,3,1	s,NIL,a	a,b,c
18	relabel-reset b	5,6,1,0,0	-8,0,4,3,1	s,s,a	a,b,c
19	advance b b→s	5,6,1,0,0	-8,0,4,3,1	s,c,a	a,b,c
20	push b b→c δ=2	5,6,1,0,0	-8,0,2,5,1	s,c,a	a,b,c

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Steps 21–46

step	operation	h(s,a,b,c,t)	e(s,a,b,c,t)	current(a,b,c)	L
21	advance b b→c	5,6,1,0,0	-8,0,2,5,1	s,NIL,a	a,b,c
22	relabel-reset b	5,6,6,0,0	-8,0,2,5,1	s,s,a	a,b,c
23	push b b→s δ=2	5,6,6,0,0	-6,0,0,5,1	s,s,a	a,b,c
24	discharge-end b	5,6,6,0,0	-6,0,0,5,1	s,s,a	a,b,c
25	move-front b	5,6,6,0,0	-6,0,0,5,1	s,s,a	b,a,c
26	discharge-start a	5,6,6,0,0	-6,0,0,5,1	s,s,a	b,a,c
27	discharge-end a	5,6,6,0,0	-6,0,0,5,1	s,s,a	b,a,c
28	discharge-start c	5,6,6,0,0	-6,0,0,5,1	s,s,a	b,a,c
29	advance c c→a	5,6,6,0,0	-6,0,0,5,1	s,s,b	b,a,c
30	advance c c→b	5,6,6,0,0	-6,0,0,5,1	s,s,t	b,a,c
31	advance c c→t	5,6,6,0,0	-6,0,0,5,1	s,s,NIL	b,a,c
32	relabel-reset c	5,6,6,1,0	-6,0,0,5,1	s,s,a	b,a,c
33	advance c c→a	5,6,6,1,0	-6,0,0,5,1	s,s,b	b,a,c
34	advance c c→b	5,6,6,1,0	-6,0,0,5,1	s,s,t	b,a,c
35	push c c→t δ=4	5,6,6,1,0	-6,0,0,1,5	s,s,t	b,a,c
36	advance c c→t	5,6,6,1,0	-6,0,0,1,5	s,s,NIL	b,a,c
37	relabel-reset c	5,6,6,7,0	-6,0,0,1,5	s,s,a	b,a,c
38	push c c→a δ=1	5,6,6,7,0	-6,1,0,0,5	s,s,a	b,a,c
39	discharge-end c	5,6,6,7,0	-6,1,0,0,5	s,s,a	b,a,c
40	move-front c	5,6,6,7,0	-6,1,0,0,5	s,s,a	c,b,a
41	discharge-start b	5,6,6,7,0	-6,1,0,0,5	s,s,a	c,b,a
42	discharge-end b	5,6,6,7,0	-6,1,0,0,5	s,s,a	c,b,a
43	discharge-start a	5,6,6,7,0	-6,1,0,0,5	s,s,a	c,b,a
44	push a a→s δ=1	5,6,6,7,0	-5,0,0,0,5	s,s,a	c,b,a
45	discharge-end a	5,6,6,7,0	-5,0,0,0,5	s,s,a	c,b,a
46	terminate	5,6,6,7,0	-5,0,0,0,5	s,s,a	c,b,a

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A09-M01

Send 1 on each of s-a-2-t, s-b-1-t, s-c-3-t. Matching {a2,b1,c3} has size 3. Cut {s} has capacity 3, proving optimality.

A09-M02

Unit terminal capacities, integrality, and conservation allow at most one selected middle arc per vertex. In $K_{2,2}$ send 1/2 on all four middle arcs and 1 on all terminal arcs. Value is 2, but all four middle arcs are positive. The theorem guarantees existence, not universality.

A09-M03

M={a1,c2}. Cut S={s,a,b,1} has outgoing arcs s→c and 1→t, totaling 2. Set A={a,b} has neighborhood {1}.

A09-M04

Use s→v_in=5, v_in→v_out=3, v_out→t=5, and send 3. Cut {s,v_in} has capacity 3. Conservation gives total inflow=internal flow=total outflow. Size is n+k vertices and m+k arcs.

A09-M05

Use quota arcs of capacities 3,2,3,2. Cut {S,a,b,v,d} has outgoing arcs v→c=2 and d→T=2, totaling 4<5. Without sink quotas, send 2 to c and 3 to d for optimum 5.

A09-M06

d=(-2,1,1), residual capacities=(2,2,3). Add S*→B=1, S*→C=1, A→T*=2. Route 1 through B-C-A and 1 through C-A. Then y=(0,1,2), x=(2,2,2). Every vertex receives and sends 2. The sign follows from out(y)−in(y)=in(ℓ)−out(ℓ).

A09-M07

D=2. Cut {S*,B,C} has only outgoing arc C→A of capacity 1. A flow of 1 exists, so the auxiliary optimum is exactly 1. At A, required outflow is at least 2 but possible inflow at most 1.

A09-M08

For F=2 add t→s:[2,2], giving d=(1,0,−1). Send 1 along S*-s-a-t-T* and recover 2 on both original arcs. F=3 violates upper bound 2 on a→t. For unspecified nonnegative value, M=3+2=5 is safe. Integral bounds permit integral auxiliary flow and integral recovery x=ℓ+y. Arbitrary real bounds do not guarantee integrality.

References and attribution

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